# CS411

# SOFTWARE ARCHITECTURE DESIGN

# SYLLABUS

# 2021-2022 Spring


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|---|---|
| Instructor: | Haluk Altunel, PhD |
| Lecture Classroom: | Section-1 (EB-204)<br>Section-2 (EB-204) |
| Lecture Hours: | Section-1 Tuesday (10:30 – 12:20) - Thursday (15:30-17:20)<br>Section-2 Tuesday (8:30 – 10:20) - Thursday (13:30-15:20) |
| Office Hours: | By appointment |
| E-Mail: | altunel@bilkent.edu.tr |
| TA: | TBD |


## Course Objectives

Basic concepts, methods and techniques for designing software architectures; rationale for software architecture design, modeling software architecture design, architectural styles/patterns, architectural requirements analysis, comparison and evaluation of architecture design methods, synthesis-based software architecture design, software product-line architectures, domain modeling, domain engineering and application engineering, software architecture implementation, evaluating software architecture designs. Current topics on microservice architecture, serverless architecture, and event-driven architecture will be covered.

## Grading

| | Item | Type | Total Contribution |
|---|---|---|---|
| 1 | Quizzes & In-class Participation | Individual | 15 |
| 2 | Paper Review | Individual | 10 |
| 3 | Final - Essay/Written/Open Book | Individual | 30 |
| 4 | Project-1 | Team | 20 |
| 5 | Project-2 | Team | 25 |
| **TOTAL** | | | **100** |

## Schedule

| TOPIC |
|---|
| Introduction + Motivation for Software Architecture Design |
| Modeling Software Architectures |
| Architectural Design Methods - Classifying architecture design methods - Architectural Requirements Analysis |
| Architectural Design Methods - Domain Analysis - Synthesis-based architecture design method |
| Architectural Views - Module Styles |
| Architectural Views - Component & Connector Styles |
| Architectural Views - Component & Connector Styles |
| Project 1: Project Presentations |
| Architectural Views - Allocation Styles - Documenting Architecture using Architecture Views |
| Architectural Views - Documenting Architecture using Architecture Views |
| Architectural Pattern |
| Architectural Evaluations |
| Software Product Line Engineering |
| Project 2: Project Presentations |

## Recommended Textbooks & References

- Required - Textbook: Documenting Software Architectures: Views and Beyond (2nd Edition), Clements, et al, 2010, Addison Wesley
- Recommended - Textbook: Software Architecture in Practice, Bass, et al., 2012, Addison Wesley
- Recommended - Textbook: Software Architect's Handbook: Become a successful software architect by implementing effective architecture concepts, Ingeno, 2018, Packt
- Recommended - Textbook: Fundamentals of Software Architecture: An Engineering Approach, Richards & Ford, 2020, O'Reilly
- Recommended - Textbook: Clean Architecture: A Craftsman's Guide to Software Structure and Design, Martin, 2017, Pearson
- ISO/IEC/IEEE 42010:2011, "Systems and software engineering – Architecture description"